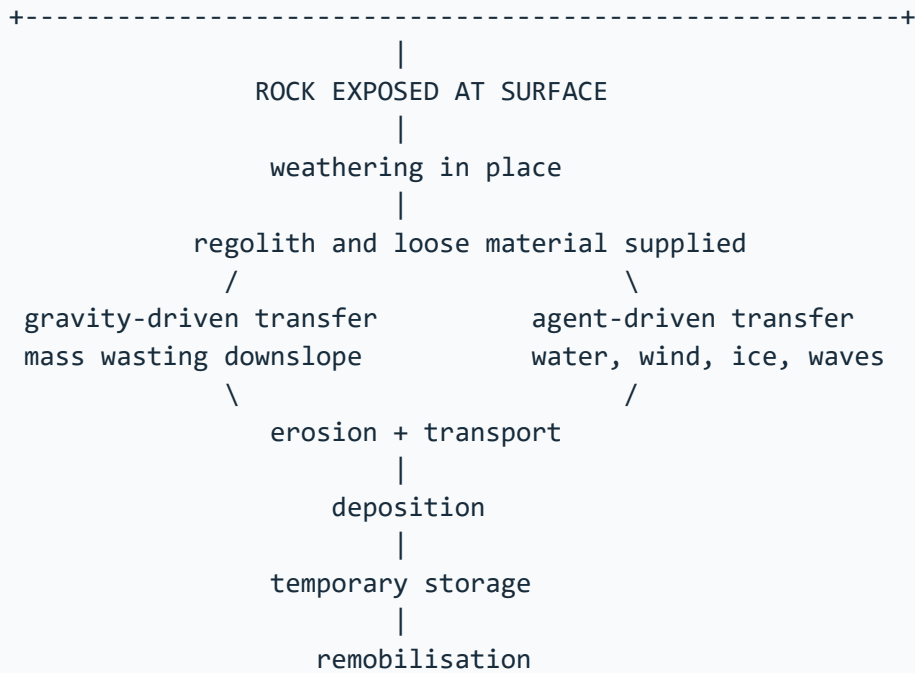


## ASCII MASTER FLOW – PANEL 1/8: Exogenic source-transfer-storage system

### EXOGENIC LANDSCAPE SYSTEM

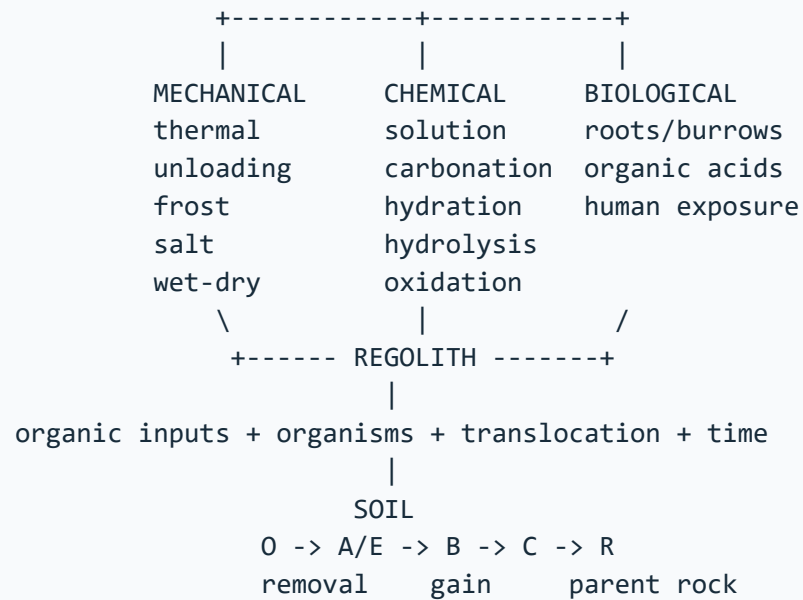
| CENTRAL QUESTION: How is rock converted into landform? |



Denudation is collective lowering; weathering, erosion and mass wasting are linked but distinct processes.

## ASCII MASTER FLOW – PANEL 2/8: Weathering taxonomy, controls, regolith and soil formation

### WEATHERING



CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief  
+ drainage + organisms + time + land use

Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks;  
parent material does not alone determine every soil property.

## ASCII MASTER FLOW – PANEL 3/8: Soil erosion sequence and watershed control logic

RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION

```
      |
diffuse sheet removal
      |
small rill channels
      |
entrenched gullies
      |
ravine landscape: Chambal cue
```

### WATERSHED RESPONSE

cover soil -> slow runoff -> shorten slope length -> contour/terrace  
-> protect drainage -> treat gully head -> rehabilitate catchment

### WIND BRANCH

surface creep -> saltation -> suspension

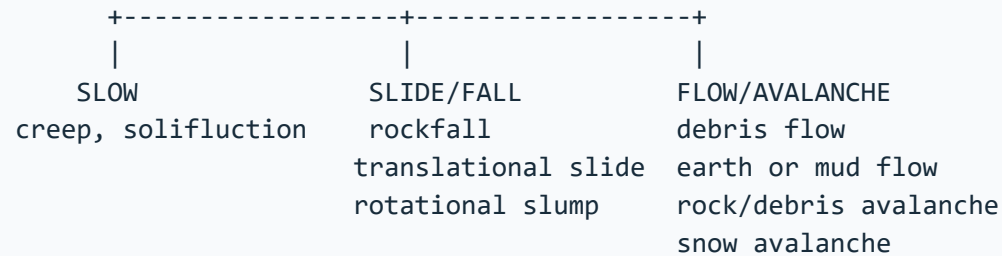
### COASTAL BRANCH

wave erosion belongs to the coastal-process system; use only when asked.

A check dam alone cannot replace upstream cover, drainage and maintenance.

## ASCII MASTER FLOW – PANEL 4/8: Mass-movement taxonomy and slope-stability mechanism

### MASS MOVEMENT



### SLOPE BALANCE

resisting strength = cohesion + friction + structural support

versus

driving stress = gravity component + loading + toe loss

rain infiltration -> added weight + pore pressure -> lower effective strength

road or river cut -> steepening / undercutting -> support loss

quake / quarry / leakage / drainage change -> trigger or amplification

FAILURE WHEN DRIVE > RESISTANCE

Material, movement and water define the event; speed alone does not.

## ASCII MASTER FLOW – PANEL 5/8: India landslide geography, risk production and mitigation stack

### INDIA LANDSLIDE GEOGRAPHY

Himalaya / NE : young relief + fractures + incision + rain + seismicity  
Western Ghats : escarpment + deep weathering + orographic rain + cuts  
Nilgiris : weathered slopes + roads + settlements + drainage change

|

PREDISPOSITION -> TRIGGER -> MOVEMENT -> EXPOSURE -> LOSS  
geology/slope    rain/quake    slide/flow    road/town    disaster

|

### GSI LADDER

inventory -> susceptibility/hazard study -> post-event diagnosis  
-> monitoring / NLFC support -> decision communication

|

### MITIGATION STACK

avoid unsafe siting -> manage drainage -> protect toe  
-> stabilise slope -> bioengineer -> control spoil/quarrying  
-> maintain works -> warn/close routes -> evacuate and prepare

Inventory maps past events; susceptibility maps relative likelihood;  
neither is a deterministic event prediction.  
Joshimath is a multi-causal subsidence case, not a synonym for landslide.

## ASCII MASTER FLOW – PANEL 6/8: Groundwater architecture from recharge to wells

rain → infiltration → soil moisture → percolation → recharge

GROUND SURFACE      recharge area      spring at slope

-----v----->

ZONE OF AERATION      perched water on local aquitard

----- capillary fringe -----

WATER TABLE      UNCONFINED AQUIFER      pumping well

~~~~~\\_\_\_\_\_ cone of depression

AQUITARD      slow transmission

-----

CONFINED AQUIFER      pressure; water rises to potentiometric level

=====

AQUICLUDE      practically no transmission

porosity = fraction of voids | permeability = connected flow capacity

aquifer stores and transmits | aquitard transmits slowly

artesian water rises under pressure; it flows only if head exceeds ground

Budget: change in storage = recharge + inflow - discharge - pumping - outflow.

## ASCII MASTER FLOW – PANEL 7/8: India groundwater depletion, quality and subsidence geography

| REGION         | AQUIFER FORM         | MAJOR PRESSURE         |
|----------------|----------------------|------------------------|
| Indo-Gangetic  | layered alluvium     | depletion, As, nitrate |
| Peninsular     | weathered fractures  | discontinuous storage  |
| Deccan basalt  | contacts + fractures | uneven well yield      |
| Arid hard rock | limited recharge     | fluoride, salinity     |
| Coastal belt   | fresh-saline wedge   | intrusion and upconing |

crop-water demand + cheap energy + private wells + urban growth  
+ weak aquifer regulation + limited recharge

|  
falling head -> lower yield -> higher lift cost -> unequal access

|  
compressible sediment: pore-pressure loss -> compaction -> subsidence

Quality paths differ: arsenic redox | fluoride rock-water interaction

nitrate sewage/fertiliser | salinity coastal or inland

Quantity and quality require separate evidence, though each limits usable water.

## ASCII MASTER FLOW – PANEL 8/8: Recharge, demand governance and integrated Mains answer spine

GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS

### SUPPLY / RECHARGE

roof and runoff capture  
managed aquifer recharge  
watershed treatment  
spring-shed restoration  
quality-safe source water  
maintenance and monitoring

### DEMAND / GOVERNANCE

crop-water alignment  
efficient irrigation  
well registration/metering  
conjunctive use and reuse  
community water budgets  
enforceable state/local rules

\  
+----- AQUIFER PLAN -----+

|  
CGWB assessment + NAQUIM mapping + state regulation  
+ Atal Bhujal learning + local institutions + public data

### ANSWER SPINE

1 define stock-flow-quality system | 2 draw aquifer or India map  
3 diagnose regional mechanism | 4 use dated evidence carefully  
5 match intervention to cause | 6 qualify limits and equity

Recharge structures are inputs, not proof of water-table recovery.  
Desalination supplies water but does not restore freshwater aquifer head.